

REQUIREMENT ANALYSIS

Sephora Beauty E-commerce Website

1. Project Introduction

The **Sephora Beauty E-commerce Website** is an online shopping platform developed to provide customers with an easy, secure, and convenient way to purchase beauty and cosmetic products. The website offers a wide range of skincare, makeup, haircare, fragrances, and beauty accessories from various premium brands.

The system allows users to browse products, search for specific items, add products to the shopping cart, place orders, make secure online payments, and track their orders. Administrators can manage products, categories, customers, and orders through an admin dashboard.

The main objective of this project is to simplify the shopping experience while providing customers with a modern, user-friendly, and reliable online platform.

2. Requirement Analysis

Requirement Analysis is the process of identifying, gathering, and documenting the needs of users and the system before development begins. It helps developers understand what features are required and ensures that the final application satisfies customer expectations.

For the Sephora Beauty E-commerce Website, the requirement analysis focuses on customer needs, product management, secure transactions, order processing, and system performance.

The requirements are divided into:

- Functional Requirements
 - Non-Functional Requirements
 - Software Requirements
 - Hardware Requirements
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3. Objectives

The main objectives of this project are:

- To provide an online platform for purchasing beauty products.
- To offer a simple and attractive user interface.
- To reduce manual shopping effort.
- To enable secure online payments.
- To allow customers to search and filter products easily.
- To manage customer orders efficiently.

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- To maintain product inventory accurately.
 - To improve customer satisfaction through quick service.
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4. Scope

The scope of the Sephora Beauty E-commerce Website includes:

- Customer Registration
- User Login
- Product Browsing
- Product Search
- Product Categories
- Shopping Cart
- Wishlist
- Secure Payment
- Order Confirmation
- Order Tracking
- Customer Profile Management
- Product Reviews and Ratings
- Admin Dashboard
- Inventory Management
- Sales Reports

Future enhancements may include:

- AI-based product recommendations
 - Virtual makeup try-on
 - Chatbot support
 - Loyalty reward points
 - Mobile application
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5. Existing System

In the existing system:

Customers usually visit physical stores to purchase beauty products.

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Problems of Existing System

- Time-consuming shopping process.
 - Limited product availability.
 - Manual billing.
 - Difficult product comparison.
 - No personalized recommendations.
 - Long waiting queues.
 - Limited shopping hours.
 - Travel expenses.
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6. Proposed System

The proposed Sephora Beauty E-commerce Website provides an online platform where customers can purchase beauty products anytime and from anywhere.

Features

- User Registration
- Secure Login
- Product Search
- Product Categories
- Shopping Cart
- Wishlist
- Online Payment
- Order Tracking
- Customer Reviews
- Admin Dashboard
- Inventory Management
- Sales Report Generation

Benefits

- Saves time
- Easy shopping
- Secure payment
- Faster order processing

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- Better customer experience
 - Easy inventory management
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7. Functional Requirements

Functional requirements describe what the system should do.

Customer Module

- Register new account
- Login securely
- Update profile
- Change password

Product Module

- View products
- Search products
- Filter by category
- View product details

Shopping Cart Module

- Add products
- Remove products
- Update quantity
- View total amount

Wishlist Module

- Save favorite products
- Remove products

Payment Module

- Online payment
- Payment confirmation
- Invoice generation

Order Module

- Place order
- Cancel order

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- Track order
- Order history

Admin Module

- Manage products
 - Manage categories
 - Manage users
 - Manage orders
 - Manage inventory
 - Generate reports
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8. Non-Functional Requirements

Performance

- Fast loading website
- Quick search results
- Smooth navigation

Security

- Secure login
- Password encryption
- Secure payment gateway
- Data privacy

Reliability

- Continuous availability
- Accurate order processing
- Data backup

Usability

- User-friendly interface
- Easy navigation
- Responsive design

Scalability

- Support thousands of users

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- Easy database expansion

Maintainability

- Easy software updates
 - Easy bug fixing
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9. Software Requirements

Software	Purpose
Windows 10/11	Operating System
Visual Studio Code	Code Editor
HTML5	Front-end
CSS3	Styling
JavaScript	Client-side scripting
React.js (Optional)	Front-end Framework
Node.js	Backend Runtime
Express.js	Backend Framework
MongoDB	Database
Git	Version Control
GitHub	Project Repository
Chrome	Testing Browser

10. Hardware Requirements

Hardware Specification

Processor	Intel Core i3 or above
RAM	4 GB Minimum
Storage	100 GB Free Space
Monitor	1366 × 768 Resolution
Keyboard	Standard Keyboard

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Hardware Specification

Mouse Optical Mouse

Internet Broadband Connection

11. Modules

Module 1 – User Management

- Registration
 - Login
 - Profile Update
 - Logout
-

Module 2 – Product Management

- Add Product
 - Edit Product
 - Delete Product
 - View Products
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Module 3 – Category Management

- Add Category
 - Update Category
 - Delete Category
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Module 4 – Shopping Cart

- Add to Cart
 - Remove Item
 - Update Quantity
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Module 5 – Wishlist

- Add Wishlist
- Remove Wishlist

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Module 6 – Payment

- Payment Gateway
- Order Confirmation
- Payment Status

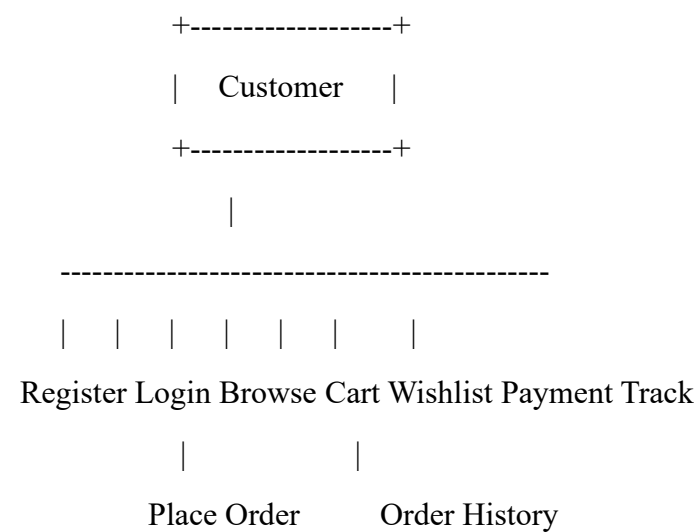
Module 7 – Order Management

- Place Order
- Cancel Order
- Order Tracking
- Order History

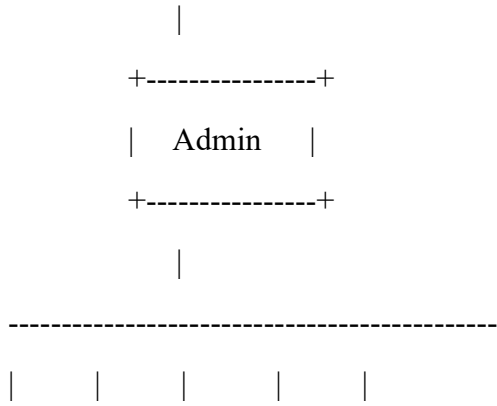
Module 8 – Admin Dashboard

- Manage Users
- Manage Products
- Manage Orders
- Inventory Management
- Sales Reports

12. Use Case Diagram



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Manage Users Manage Products Manage Orders Reports Inventory

13. ER Diagram

Customer

Customer_ID (PK)

Name

Email

Phone

Password

|

| Places

|

Orders

Order_ID (PK)

Customer_ID (FK)

Order_Date

Total_Amount

Status

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|

| Contains

|

Order_Items

Order_Item_ID (PK)

Order_ID (FK)

Product_ID (FK)

Quantity

Price

|

| References

|

Products

Product_ID (PK)

Product_Name

Category_ID (FK)

Brand

Price

Stock

|

|

Categories

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Category_ID (PK)

Category_Name

|

|

Payments

Payment_ID (PK)

Order_ID (FK)

Payment_Method

Payment_Status

14. Database Design

Customer Table

Field	Type
Customer_ID	INT
Name	VARCHAR
Email	VARCHAR
Phone	VARCHAR
Password	VARCHAR

Product Table

Field	Type
Product_ID	INT
Product_Name	VARCHAR
Brand	VARCHAR
Price	DECIMAL

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Field	Type
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Stock	INT
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Category_ID	INT
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Category Table

Field	Type
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Category_ID	INT
-------------	-----

Category_Name	VARCHAR
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Orders Table

Field	Type
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Order_ID	INT
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Customer_ID	INT
-------------	-----

Order_Date	DATE
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Total_Amount	DECIMAL
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Status	VARCHAR
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Order_Items Table

Field	Type
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Order_Item_ID	INT
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Order_ID	INT
----------	-----

Product_ID	INT
------------	-----

Quantity	INT
----------	-----

Price	DECIMAL
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Payments Table

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Field	Type
Payment_ID	INT
Order_ID	INT
Payment_Method	VARCHAR
Payment_Status	VARCHAR

15. Conclusion

The **Sephora Beauty E-commerce Website** is designed to provide customers with a fast, secure, and convenient online shopping experience for beauty products. It simplifies product browsing, purchasing, payment, and order tracking while allowing administrators to efficiently manage products, customers, inventory, and sales. The system improves shopping convenience, enhances customer satisfaction, and supports business growth. Future enhancements such as AI-powered recommendations, virtual makeup try-on, and mobile app integration can further improve the platform and provide a richer user experience.